

MD ABU ZAFAR AL MUNSUR

PERSONAL INFORMATIONS

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JOB EXPERIENCE

April' 2022- present **Postdoctoral Fellow, Korea Institute of Energy Technology (KENTECH)** – Institute of Hydrogen Energy, college of natural science, Incheon, South Korea

March' 2021- February' 2022 **Presidential Postdoctoral Fellow, Incheon National University** – Department of chemistry, college of natural science, Incheon, South Korea

EDUCATION

March' 2016-February' 2021 **Incheon National University** – Incheon, South Korea
Doctor of Philosophy in Chemistry; Major in Organic Materials Synthesis
Dissertation: *Development of ion-conducting polymers with enhanced properties by surface-coating and molecular control strategies.*
Supervisor: Prof. Tae-Hyun Kim

July' 2013-February' 2016 **University of Rajshahi** – Rajshahi, Bangladesh
Master of Science in Chemistry; Major in Synthetic Organic Chemistry
Dissertation: *Highly efficient and environmentally benign synthesis of tri- and tetrasubstituted imidazole catalyzed by 3-picolinic acid (Niacin).*
Supervisor: Prof. Harendra Nath Roy

January' 2008-June' 2013 **University of Rajshahi** – Rajshahi, Bangladesh
Bachelor of Science in Chemistry; Minor in Physics and Math

RESEARCH AREA

Synthesis Design, synthesis and characterization of target-based micro and macromolecules

Fuel Cells Development of a highly efficient and robust polymer electrolyte membrane for Proton/Anion Exchange Membrane, Membrane Electrode Assembly (MEA), studying Low and HT-PEMFC/AEMFC Performances.

Electrolysis Manufacturing of low-cost and well-performed polymer electrolyte membrane for water electrolysis to hydrogen generation, Preparation of MEA, Studying Electrolysis performance.

CCS Development of energy-efficient robust polymeric membrane for carbon-dioxide separation and sequestration.

RESEARCH INTEREST

Microbial Fuel Cells	Development of foul resistant, highly efficient, and robust polymer electrolyte Membrane for MFCs.
Nano-Filtration	Manufacturing of low-cost and energy-efficient polymeric membrane for the nano-filtration process.
Solar cells	Development of opt-electric organic solar cells.

PUBLICATIONS AND PAPERS

LIST OF PATENTS

6. *Carbon quantum dot-coated polymer electrolyte membranes and method thereof*, Patent number KR102867798, Issued April 24, 2023, T.-H. Kim, K. Min, S. Jeon, **A. Z. A. Munsur**. Doi.org/10.8080/1020230053405
5. *Anion exchange membranes comprising hydrophilic-hydrophobic comb-type SEBS and method thereof*, Patent number KR102600485, Issued November 08, 2023, T.-H. Kim, **A. Z. A. Munsur**, J.-H. Lee. Doi.org/10.8080/1020210103457
4. *Nafion-based composite membrane for a proton exchange membrane, a method for manufacturing the same, a proton exchange membrane comprising the composite membrane, a fuel cell, and water electrolyzer comprising the proton exchange membrane*, Issued November 03, 2023; Patent number KR102597827, T.-H. Kim, **A. Z. A. Munsur**, B.-H. Goo. Doi.org/10.8080/1020210046641
3. *Sulfonated polyethersulfone copolymer, composite membranes for ion exchange membrane comprising the same, ion exchange membrane comprising the composite membrane, fuel cell comprising the ion exchange membrane, method for manufacturing the copolymer and method for manufacturing composite membrane for the ion exchange membrane*, Patent number KR102418820, Issued July 11, 2022, T.-H. Kim, **A. Z. A. Munsur**, W. Choi, B.-H. Goo. Doi.org/10.8080/1020200090386
2. *Anion exchange membrane and method for preparing the same*, Patent number KR102217838, Issued February 22, 2021, T.-H. Kim, S. Y. Nam, **A. Z. A. Munsur**, S. H. Sung. Doi.org/10.8080/1020180100233
1. *Method of Increasing Solubility of Polyvinyl alcohol*, Patent number KR102030501, Issued October 10, 2019, T.-H. Kim, I. Hossain, **A. Z. A. Munsur**, H.-S. Lee. Doi.org/10.8080/1020180111689

PEER-REVIEWED PUBLICATIONS (*DENOTES EQUAL CONTRIBUTION AND *DENOTES CORRESPONDING AUTHOR)

UNDER PREPARATION:

3. M. Rana[†], **A. Z. A. Munsur**[†], J. Han, H. J. Kim* "A comprehensive study on URFC: The state of the art, remaining challenges, and the prospects" manuscript under preparation
2. M. Aftabuzzaman[†], **A. Z. A. Munsur**[†], J. Han, H. J. Kim* "Nanostructured Carbons and Their Composites for Energy Storage and Conversion Devices" manuscript under preparation
1. **A. Z. A. Munsur**, H. Cho, B. N. Lee, M. Rana, J. Han, H. J. Kim* "Polybenzimidazole-Based in Situ Thermally Induced Semi-Interpenetrating Polymer Electrolyte Membranes" manuscript under preparation.

JOURNAL ARTICLES:

14. T. K. Lee[†], Min, J. Kim[†], H. Cho[†], S. Lee[†], J. Kim, **A. Z. A. Munsur**, B.-N. Lee, J. E. Chae, J. Han, H. S. Park, J. H. Jang, S. J. Yoo, S. Y. Peak, S. Y. Lee,^{*} H.-S. Cho,^{*} S.-Y. Park,^{*} K. H. Song,^{*} H.-J. Kim,^{*} Long-Term, Stable Alkaline Electrolysis with a Durable Crystalline Polybenzimidazole Membrane. **Small** 21, no. 38 (2025): e02810. Doi.org/10.1002/sml.202502810
13. K. Min[†], **A. Z. A. Munsur**[†], S. Y. Paek, S. Jeon, S. Y. Lee, and T.-H. Kim,^{*} Development of High-performance Polymer Electrolyte Membranes through the Application of Quantum Dot Coatings to Nafion Membranes. **ACS Appl. Mater. Interfaces** 2023, 15, 15616-15624. DOI:10.1021/acsami.3c01289
12. **A. Zafar Al Munsur**, J. Lee, J. E. Chae, H.-J. Kim, C. H. Park, S. Y. Nam,^{*} and T.-H. Kim,^{*} Hexyl quaternary ammonium- and fluorobenzoyl-grafted SEBS polymers as hydrophilic-hydrophobic comb-type anion exchange membranes. **J. Membr. Sci.** 2022, 643, 120029. DOI:10.1016/j.memsci.2021.120029
11. B.-H. Goo, S. Y. Paek, **A. Z. A. Munsur**, O. Choi, Y. Kim, O. J. Kwon, S. Y. Lee, H.-J. Kim, and T.-H. Kim,^{*} Polyamide-coated Nafion composite membranes with reduced hydrogen crossover produced via interfacial polymerization. **Int. J. Hydrog. Energy** 2021, 47, 1202-1216. DOI:10.1016/j.ijhydene.2021.10.074
10. **A. Z. A. Munsur**, B.-H. Goo, Y. Kim, O. J. Kwon, S. Y. Paek, S. Y. Lee, H.-J. Kim, and T.-H. Kim,^{*} Nafion-based proton exchange membranes with a cross-linked semi-interpenetrating polymer network between PVA and PAA. **ACS appl. Mater. interfaces** 2021, 13 (24), 28188-28200. DOI: 10.1021/acsami.1c05662
9. B.-H. Goo[†], **A. Z. A. Munsur**[†], O. Choi, Y.-K. Kim, O.-J. Kwon, and T.-H. Kim^{*} Sulfonated Poly(Ether Sulfone)-Coated and Blended Nafion Membranes with Enhanced Conductivity and Reduced Hydrogen Permeability. **ACS Appl. Energy Mater.** 2020, 3, 11418-11433. DOI: 10.1021/acsaem.0c02319
8. **A. Z. A. Munsur**,^{*} H. N. Roy, and M. Khan, Highly Efficient and metal-Free Synthesis of Try- and Tetrasubstituted Imidazole Catalyzed by 3-Picolinic Acid. **Arab. J. Chem.** 2020, 13, 8807-8814. DOI: 10.1016/j.arabjc.2020.10.010
7. I. Hossain, D. Kim, **A. Z. A. Munsur**, J. M. Roh, H. B. Park, and T.-H. Kim,^{*} PEG/PPG-PDMS-based Crosslinked Copolymer Membranes Prepared by ROMP and in situ Membrane Casting for CO₂ Separation: an approach to endow rubbery materials with properties of rigid polymers. **ACS Appl. Mater. Interfaces** 2020, 12(24), 27286–27299. DOI: 10.1021/acsami.0c06926
6. **A. Z. A. Munsur**, I. Hossain, S. Y. Nam, J. E. Chae, and T.-H. Kim,^{*} Quaternary ammonium-functionalized hexyl bis (quaternary ammonium)-mediated partially crosslinked SEBSs as highly conductive and stable anion exchange membranes. **Int. J. Hydrog. Energy** 2020, 45(31), 15658-15671. DOI: 10.1016/j.ijhydene.2020.04.063
5. **A. Z. A. Munsur**, I. Hossain, S. Y. Nam, J. E. Chae, and T.-H. Kim,^{*} Hydrophobic-hydrophilic comb-type quaternary ammonium-functionalized SEBS copolymers for high performance anion exchange membranes. **J. Membr. Sci.** 2020, 599, 117829. DOI:10.1016/j.memsci.2020.117829
4. I. Hossain, **A. Z. A. Munsur**, T.-H. Kim,^{*} A Facile Synthesis of (PIM-Polyimide)- (6FDADurene-Polyimide) Copolymer as Novel Polymer Membranes for CO₂ Separation. **Membranes** 2019, 9, 113(1-14). DOI:10.3390/membranes9090113

3. I. Hossain, **A. Z. A. Munsur**, O. Choi, and T.-H. Kim,* *Bisimidazolium PEG-mediated crosslinked 6FDA-durene polyimide membranes for CO₂ separation.* **Sep. Purif. Technol.** 2019, 224, 690-697. DOI: 10.1016/j.seppur.2019.05.014
2. H. Lim, B. Lee, D. Yun, **A. Z. A. Munsur**, J. E. Chae, S. Y. Lee, H.-J. Kim, S. Y. Nam, C.-H. Park, and T.-H. Kim,* *Poly(2,6-dimethyl-1,4-phenylene oxide)s with various head groups: effect of head groups on the properties of anion exchange membranes.* **ACS Appl. Mater. Interfaces** 2018, 10(48), 41279-41292. DOI:10.1021/acsami.8b13016
1. H. N. Roy,* , M. Rana, **A. Z. A. Munsur**, K.-I. Lee & A. K. Sarker, *Efficient and convenient synthesis of 1H-pyrazolo[1,2-b] phthalazine-5,10-dione derivatives mediated by L-proline.* **Synth. Commun.** 2016, 46 (16), 1370-1376. DOI:10.1080/00397911.2016.1192650 (Selected Abstracted in **ChemInform**)

CONFERENCE PROCEEDINGS & PRESENTATIONS

16. **A. Z. A. Munsur**, H. Cho, Ms. B.-N. Lee, H.-J. Kim* "Polybenzimidazole-based polymer electrolyte membranes form an in-situ thermally induced semi-interpenetrating polymer network between PVA and PAA" [**Symposium 06**] **Advanced Materials and Devices for Fuel Cell and Electrolysis Technologies, ICAE 2023, 7th International Conference on Advanced Electromaterials, October 31 (Tue.)-November 3 (Fri.), 2023, Ramada Plaza Jeju Hotel, Jeju, South Korea, (Poster).**
15. B.-N. Lee, H. Cho, **A. Z. A. Munsur**, H.-J. Kim* "The Effect of Temperature and Electrolyte Concentration on The AEMWE Performance with p-Polybenzimidazole Membrane" [**Symposium 06**] **Advanced Materials and Devices for Fuel Cell and Electrolysis Technologies, ICAE 2023, 7th International Conference on Advanced Electromaterials, October 31 (Tue.)-November 3 (Fri.), 2023, Ramada Plaza Jeju Hotel, Jeju, South Korea, (Poster).**
14. H. Cho, B.-N. Lee, **A. Z. A. Munsur**, H.-J. Kim* "Exploring the Effectiveness of Alkaline Ion-Solvating Water Electrolysis Copolymer Membranes in the Polybenzimidazole Family Groups: Delving into Enhanced Performance and Emerging Applications" [**Symposium 06**] **Advanced Materials and Devices for Fuel Cell and Electrolysis Technologies, ICAE 2023, 7th International Conference on Advanced Electromaterials, October 31 (Tue.)-November 3 (Fri.), 2023, Ramada Plaza Jeju Hotel, Jeju, South Korea, (Poster).**
13. **A. Z. Al Munsur**, T.-H. Kim* "Nafion-based proton exchange membranes with a semi-interpenetrating polymer network between PVA and PAA" **Korean Electrochemical Society meeting, 2021 spring General Assembly and academic Presentation, April 8(Thus.) ~ 10(Sat.), 2021, BEXCO, Busan, South Korea, (Poster).**
12. **A. Z. A. Munsur**, T.-H. Kim* "A systematic study of the effect of hydrophobic side chain on the properties of poly(styrene-*b*-ethylene-co-butylene-*b*-styrene) triblock copolymer-based anion exchange membranes" **2020 Korea Membrane Society Fall Meeting (30th Anniversary), November 18-20, 2020, Hotel ICC Daejeon & Online, South Korea, (Poster).**
11. **A. Z. A. Munsur**, T.-H. Kim* "A Molecular Approach to Control the Morphology and Properties for the Development of Highly Conductive and Stable Anion Exchange Membranes" **PRiME 2020 (ECS, ECSJ, & KECS Joint Meeting), (Oral).**
10. B.-H. Goo, **A. Z. A. Munsur**, O. Choi, Y. K. Kim, O. J. Kwon, T.-H. Kim* "Development of Nafion Coating and Blending Membrane of Sulfonated PES Polymer for Improved Performance and Reduced Hydrogen Permeability" **The Membrane Society of Korea, 2020 Summer Conference and workshop, 20-22nd August 2020, South Korea, (Poster).**
9. **A. Z. A. Munsur**, T.-H. Kim* "Convenient route to develop highly conductive and alkaline stable anion exchange membranes for alkaline fuel cell applications." **The Korean Ceramic Society, the 22nd International Conference on Solid State Ionics (SSI-22), June 16 to 21, 2019, PyeongChang, Korea, (Poster).**

8. **A. Z. A. Munsur, T.-H. Kim*** "Tuning the cross-linker lengths and cross-linking processes for the development of robust anion exchange membranes" **Korea Polymer Society, Fall Conference Oct. 10, 2018 (Wed) - 12 (Fri) at Gyeongju Agricultural Convention Center (HICO), South Korea, (Poster).**
7. **A. Z. A. Munsur, T.-H. Kim*** "Tuning of hydrophobic side-chain poly(styrene-ethylene-co-butylene-styrene) tri-block copolymer: a comparative study of free-volume effect on anion exchange membranes" **The Membrane Society of Korea, 2018 Spring General Assembly and Conference, May 2018 May 17 (Thu) ~ Fri the 18th, Korea Institute of Science and Technology, Seoul, (Poster).**
6. **A. Z. A. Munsur, T.-H. Kim*** "Tunable hydrophobic side-chain poly(styrene-ethylene-co-butylene-styrene) triblock copolymer as highly conductive and stable anion exchange membranes" **The Korean Society of Industrial and Engineering Chemistry, 2018 Spring General Assembly and Conference held on Wed 2 May - Fri, May 4, Daegu EXCO, South Korea, (Poster).**
5. **A. Z. A. Munsur, T.-H. Kim*** "Partially cross-linked poly(styrene-ethylene-co-butylene-styrene) ionomers as highly conductive and stable anion exchange membrane" **The Korean Society of Industrial and Engineering Chemistry, 2017 fall meeting November 8(Wed.) ~ 10(Fri), 2017, BEXCO, Busan, South Korea, (Poster).**
4. **A. Z. A. Munsur, T.-H. Kim*** "Crosslinked poly(styrene-ethylene-co-butylene-styrene) ionomers as highly conductive and stable anion exchange membranes" **The Korean Society of Industrial and Engineering Chemistry, 2017 Spring General Assembly and Conference, May 10, 2017 (Wed.) ~ Fri, May 12, Kim Dae-jung Convention Center, Gwangju, South Korea.**
3. **A. Z. A. Munsur, T.-H. Kim*** 다양한 사슬 길이를 갖는 SEBS 공중합체 기반의 음이온 교환막 (Tuning of hydrophobic side-chain poly(styrene-ethylene-co-butylene-styrene) tri-block copolymer: a comparative study of free-volume effect on anion exchange membranes). **The Membrane Society of Korea, 2018 Spring General Assembly, and Conference, May 2018 May 17 (Thu) ~ Fri the 18th, Korea Institute of Science and Technology, Seoul. (An elevator pitch/ Fast talk), (Oral).**
2. **A. Z. A. Munsur, T.-H. Kim*** The effect of hydrophobic side chain on the properties of Alkaline Anion Exchange Membrane Fuel Cells, **The Korean Society of Industrial and Engineering Chemistry 2021 Fall General Assembly and Conference held on Wed 4 Nov - Fri, Nov 5, Daegu EXCO, (Oral).**
1. **A. Z. A. Munsur, T.-H. Kim*** 소수성 빗살형 측쇄가 알칼리 음이온 교환막 연료전지 특성에 미치는 영향 (The effect of hydrophobic-comb type side chain on the properties of Alkaline Anion Exchange Membrane Fuel Cells). **The Membrane Society of Korea 2021 Fall General Assembly and Conference, Nov 2021 Nov 18(thus)-19 (Fri), Ramada Hotel, Daejeon, (Oral).**

INVITED TALKS

1. **Hydrogen is an energy vector for next generation energy solutions**, Research seminar for graduate/post-graduate students and faculty, University of Rajshahi, Rajshahi-6205, Bangladesh, Oct. 20, 2025.

AWARDS

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|-------------|---|
| 2019 | Graduate research excellence award (By the Institute of Basic Science, Incheon National University) |
| 2021 | Excellent oral presentation award (By the Membrane Society of Korea) |

FELLOWSHIPS AND GRANTS

6. Incheon National University Presidential Postdoctoral Research Assistance Program (2021), South Korea.
5. National University Research, Science and Technology grant for the recycling of marine resources based on the blue carbon biomass Graduate research excellence award (By the Institute of Basic Science, Incheon National University), South Korea
4. Incheon National University Research Grant for Graduate Students, South Korea
3. The Basic Science Research Program through the National Research Foundation Korea (NRF) funded by the Ministry of Education (NRF-2017R1A6A1A06015181), South Korea

2. National Research Foundation of Korea (NRF) grant was funded by the Ministry of Science, ICT & Future Planning (NRF-2015M1A2A2058013), South Korea
1. 100% Tuition fees waived fellowships were awarded by Incheon National University, South Korea

SKILLS

Synthetic Skills

Experienced in

Metathesis (ROMP, RCM, and cross-metathesis) reaction using Grubbs' 2nd Gen. & 3rd Gen. catalysts.
 Freeze-Pump-Thaw degassing process of liquid and use of canula.
 Friedel-craft polymerization and acylation reaction.
 Polycondensation reaction for making block and random copolymers.
 Synthesis of PBI and ABPBI Polymers.
 Synthesis of polymer in intrinsic microporosity.
 Using autoclave for hydrothermal and solvothermal reactions.
 Using of the pressured vessel for high-temperature Diels-Alder reactions
 Synthesis of heterocyclic carbenes using organo-catalyst, purification, and separation.
 Control polymerization e.g., ATRP, RAFT
 Catalyst slurry preparation and MEA formation (CCM, CCS, and decal method)

Analytical Skills

Experienced in the elucidation of TGA, DMA, DSC, IR, NMR, XRD, XPS, TEM, SEM, AFM, and EIS spectra.

Instrumental Skills

Experienced in

Operating those instruments; NMR, TGA, DSC, AFM, UTM, UV, IR, EIS (Biologic), GPC (Agilent Technology), Contact angle MPLC, centrifuge Machine, Spin-coating, Doctor blade-coating, Spray coating, auto-spray coating, Fuel cell and Electrolysis station operation (CNL, South Korea).

Language Proficiency

Bengali Both Verbal and writing (Mother Tongue)
 English Fluent in both Verbal and writing.

Computer Skill

Application Software

OriginPro, SigmaPlot, ACDLABS12, CS Chem-Office, ChemDraw Ultra 12.0, Adobe Photoshop, and Mendeley, .

Internet

Sufficient Skills at E-mail, Web Searching, Sci-Finder, Reaxys, Internet browsing & social Networking.

Mentoring skills

A couple of dozen undergraduate and graduate students were trained in basic experiments for organic material synthesis in the laboratory.

Teaching Skills

A couple of O-level, A-level, and junior undergraduate students were tutored in basic/advanced organic chemistry.

RELATED EXPERIENCE

Industry-University collaborative project

2018 – 2019 Development of highly water-soluble polyvinyl alcohol (Incheon National University-Seohan Antamine Co., Ltd.)

Leadership and Volunteering

2018- 2020 ET-RT team (As a volunteer member of Bangladeshi Students' Association in Korea, www.bsak.org)

2013-2015 General Secretary at 'Shibganj Upazila (Bogura) teachers & student association, University of Rajshahi (UTSAS)'

2019-202 Advisor at 'University Student's Welfare Association, Shibjanj, Bogura (USWAS)

SERVICE AS PEER-REVIEWER

ACS Applied materials and Interfaces, ACS Applied Energy Materials, ACS Applied Polymer Materials, ACS Omega (ACS)
Advanced Materials (Wiley)
Journal of Material Chemistry A, and RSC Advanced (RSC)
International Journal of Hydrogen Energy, Journal of Power Sources, Journal of Cleaner Production, Materials Chemistry and Physics, Arabian Journal of Chemistry, Journal of Energy Storage, Renewable & Sustainable Energy Reviews, Fuel, Material Science and Engineering: B, cleaner Energy System, Chemical Engineering Journal, Results in Engineering (Elsevier)
Heterocyclic Communications (DE GRUYTER)
Polymer Bulletin, Materials for Renewable and Sustainable Energy (Springer Nature)
Engineering Research Express, Nanotechnology, Journal of Physics: Energy (IOP)
Polymers and Polymer Composites (SAGE)

MEMBERSHIPS

Student member The Korean Society of Industrial and Engineering Chemistry
The Membrane Society of Korea
The Korean Ceramic Society
Korean Polymer Society
Korean Electrochemical Society

REFERENCE

Hyoungh-Juhn Kim, Ph.D (Case Western Reserve University, USA)

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